## Magnetic hexapole model:

As the magnetic flux $\Phi$ is permeated through the magnetic pole to the pole tip when current being applied to its actuating coil, a point magnetic charge associated with the magnetic flux can therefore be defined in analogy to the electric charge as $q = \Phi/\mu_o$, where $\mu_o$ is the magnetic permeability of vacuum. The resulting magnetic field produced by the hexapole actuating system can be obtained by summing the field associated with each of the six electromagnetic poles. In other words, the resulting magnetic field in the workspace can be modeled as,

$$\mathbf{B}(\mathbf{p}) = \sum_{i=1}^{6} k_m \left(\frac{q_i}{r_i^2(\mathbf{p})}\right) \hat{\mathbf{r}}_i(\mathbf{p}),$$

where $\mathbf{p} = [x, y, x]^T$ is any spatial point in the workspace, $k_m = \mu_o/4\pi = 1.0 \times 10^{-7}\ \mathrm{N/A^2}$, $\mathbf{r}_i(\mathbf{p}) = \mathbf{p} - \mathbf{p}_{c_i}$ the displacement vector pointing from the magnetic charge $q_i$ at the position $\mathbf{p}_{c_i}$ to the spatial point $\mathbf{p}$, $r_i = \|\mathbf{r}_i\|$, and $\hat{\mathbf{r}}_i = \mathbf{r}_i/r_i$ the unit directional vector of $\mathbf{r}_i$.

## Hopkinson's law:

The magnetic flux is determined by the magnetomotive force and the reluctance of the air according to Hopkinson's law: $\boldsymbol{\Phi} = [\Phi_1, \Phi_2, \Phi_3, \Phi_4, \Phi_5, \Phi_6]^T = (N_c/\mathcal{R}_a)\mathbf{K}_I[I_1\ I_2\ I_3\ I_4\ I_5\ I_6]^T$, wherein the magnetomotive force is proportional to the input current $I_i$ and the turns of the coil $N_c$, $\mathcal{R}_a$ is the lumped magnetic reluctance from the pole tip to the workspace center in the air, and $\mathbf{K}_I$ is the 6×6 flux distribution matrix describing the magnetic coupling among 6 poles since they are connected by a magnetic yoke. The magnetic charges $\mathbf{Q} = [q_1\ q_2\ q_3\ q_4\ q_5\ q_6]^T$ can thus be related to the input currents, $\mathbf{I} = [I_1\ I_2\ I_3\ I_4\ I_5\ I_6]^T$, as described by the following equation,

$$\mathbf{Q} = \frac{\boldsymbol{\Phi}}{\mu_0} = \frac{N_c}{\mu_0 \Re_a}\mathbf{K}_I \mathbf{I}.$$

## Calibration using FEM modeling:

Let $\widehat{\mathbf{K}}_I = (6/5)k_{11}\widehat{\mathbf{K}}_I^{FEM}$ and

$$\widehat{\mathbf{K}}_I^{FEM} = \begin{bmatrix} 0.8333 & \hat{k}_{12} & \hat{k} & \hat{k} & \hat{k} & \hat{k} \\ \hat{k} & \hat{k} & \hat{k} & \hat{k} & \hat{k} & \hat{k} \\ \hat{k} & \hat{k} & \hat{k} & \hat{k} & \hat{k} & \hat{k} \\ \hat{k} & \hat{k} & \hat{k} & \hat{k} & \hat{k} & \hat{k} \\ \hat{k} & \hat{k} & \hat{k} & \hat{k} & \hat{k} & \hat{k} \\ \hat{k} & \hat{k} & \hat{k} & \hat{k} & \hat{k} & \hat{k} \end{bmatrix},$$

The estimated magnetic charges $\hat{q}_i$: $\hat{q}_i = \left(\frac{N_c}{\mu_0 \Re_a}\right)\left(\frac{6}{5}\right) k_{11} \sum_{j=1}^{6} \hat{k}_{ij} I_j$.

The resulting magnetic field: $\mathbf{B}(\mathbf{p}) = \left(\frac{N_c}{\mu_0 \Re_a}\right)\left(\frac{6}{5}\right) k_{11} \frac{\mu_0}{4\pi} \sum_{i=1}^{6}\left(\sum_{j=1}^{6} \hat{k}_{ij} I_j\right)\left(\frac{\hat{\mathbf{r}}_i(\mathbf{p})}{r_i^2(\mathbf{p})}\right)$

The estimated charge position: $\mathbf{p}_c = [\mathbf{p}_{c1} \quad \mathbf{p}_{c2} \quad \mathbf{p}_{c3} \quad \mathbf{p}_{c4} \quad \mathbf{p}_{c5} \quad \mathbf{p}_{c6}] = \hat{\ell}\begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix}$

**Spatial normalization**: Let $\overline{\mathbf{p}} = \mathbf{p}/\hat{\ell}$ and $\overline{\mathbf{p}}_c = [\overline{\mathbf{p}}_{c1} \quad \overline{\mathbf{p}}_{c2} \quad \overline{\mathbf{p}}_{c3} \quad \overline{\mathbf{p}}_{c4} \quad \overline{\mathbf{p}}_{c5} \quad \overline{\mathbf{p}}_{c6}] = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix}$

$$\mathbf{r}_i(\mathbf{p}) = \hat{\ell}\overline{\mathbf{r}}_i(\overline{\mathbf{p}}) = \hat{\ell}(\overline{\mathbf{p}} - \overline{\mathbf{p}}_{c_i})$$
$$r_i(\mathbf{p}) = \|\mathbf{r}_i(\mathbf{p})\| = \hat{\ell}\overline{r}_i(\overline{\mathbf{p}}) = \hat{\ell}\|\overline{\mathbf{p}} - \overline{\mathbf{p}}_{c_i}\|$$
$$r_i^2(\mathbf{p}) = \hat{\ell}^2\|\overline{\mathbf{p}} - \overline{\mathbf{p}}_{c_i}\|^2$$
$$\hat{\mathbf{r}}_i = \mathbf{r}_i/r_i = (\overline{\mathbf{p}} - \overline{\mathbf{p}}_{c_i})/\|\overline{\mathbf{p}} - \overline{\mathbf{p}}_{c_i}\|$$

**The resulting magnetic field**:

$$\mathbf{B}(\mathbf{p}) = \underbrace{\left(\frac{6}{5}\right)\left(\frac{N_c \cdot k_{11}}{4\pi\Re_a\hat{\ell}^2}\right)}_{\hat{g}_B}\sum_{i=1}^{6}\left(\sum_{j=1}^{6}\hat{k}_{ij}\, I_j\right)\left(\frac{\mathbf{p}/\hat{\ell} - \overline{\mathbf{p}}_{c_i}}{\|\mathbf{p}/\hat{\ell} - \overline{\mathbf{p}}_{c_i}\|^3}\right)$$

**Model parameter estimation**:

$$\varepsilon_k(\widehat{\mathbf{K}}_I^{FEM}, \hat{\ell}, \hat{g}_B) = \mathbf{B}^{FEM}(\mathbf{p}_k) - \hat{g}_B\sum_{i=1}^{6}\left(\sum_{j=1}^{6}\hat{k}_{ij}\, I_j\right)\left(\frac{\mathbf{p}/\hat{\ell} - \overline{\mathbf{p}}_{c_i}}{\|\mathbf{p}/\hat{\ell} - \overline{\mathbf{p}}_{c_i}\|^3}\right)$$

$$J(\widehat{\mathbf{K}}_I^{FEM}, \hat{\ell}, \hat{g}_B) = \sum_{k=1}^{N}\varepsilon_k^2(\widehat{\mathbf{K}}_I^{FEM}, \hat{\ell}, \hat{g}_B)$$

$$Minimize\ J(\widehat{\mathbf{K}}_I^{FEM}, \hat{\ell}, \hat{g}_B) \Longrightarrow \widehat{\mathbf{K}}_I^{FEM}, \hat{\ell}, \hat{g}_B$$